

LAB REPORT 6

MECHATRONICS SYSTEM INTEGRATION

DR. ZULKIFLI BIN ZAINAL ABIDIN

SECTION 1

GROUP 6

MATRIC NUMBER	NAME
2312369	MUHAMMAD FAHIM BIN AHMAD NORISHAM
2312479	MUHAMMAD HAZIMUDDIN BIN FAIRUZ
2312451	MUHAMMAD ZAMARUL AZIM BIN ZULKHIBRI

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WEEK 7: PLC INTERFACING WITH MICROCONTROLLER AND PC OVER ETHERNET/IP (VER 2.0)

Abstract

This lab experiment introduces us to the basic concept of PLC (Programmable Logic Controller) control using the OpenPLC editor together with an Arduino microcontroller. The main objective of this task is to design, compile, simulate, and upload a **Start and Stop Control Circuit** using ladder logic. The circuit uses two push buttons one for “Start” and another for “Stop” and an LED that behaves like a typical actuator in an industrial setup. The LED should turn ON when the Start button is pressed and stay ON until the Stop button is pressed.

This setup is one of the most common circuits used in real industrial machines such as motors, conveyors, pumps, and safety systems. By learning how to create this using OpenPLC and Arduino, we better understand how PLC logic works in real scenarios.

The experiment also teaches how to assign variables, use physical addressing for Arduino pins, connect hardware on a breadboard, and test the system to ensure the correct logic operation. This lab gives us foundational knowledge for bigger mechatronics projects that combine electronics, logic control, and microcontroller systems.

Keywords: OpenPLC, Arduino, Ladder Diagram, Start and Stop Circuit, Push Button, Mechatronics

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WEEK 7: PLC INTERFACING WITH MICROCONTROLLER & PC OVER ETHERNET/IP

1. Introduction

In modern industrial automation, Programmable Logic Controllers (PLCs) are widely used to control machines and processes due to their reliability, simple programming structure, and ease of troubleshooting. One of the most fundamental and commonly applied PLC circuits is the Start–Stop Control Circuit, which is used to operate motors, pumps, compressors, conveyor systems, and various industrial equipment.

During this lab session, the OpenPLC software served as the platform for creating the ladder diagram, which was then uploaded to an Arduino board functioning as the PLC hardware. The laptop operated as the programming station, and two push buttons acted as digital inputs: one to activate the LED output and another to deactivate it. This basic control logic demonstrates the concepts of latching and unlatching, which are essential in PLC operation.

The procedure included creating a new project in OpenPLC Editor, assigning variables, configuring pin addresses such as `%IX0.0` for inputs and `%QX0.0` for outputs, compiling the program, and running simulations before transferring the code to the Arduino. A physical circuit was assembled on the breadboard to verify the actual behavior of the inputs and outputs, ensuring that the software logic matched real hardware performance.



Figure 1.0 Shows the website to download OpenPLC at <https://autonomylogic.com/>

Our main objectives in this experiment were:

- To understand the basic operation of PLC ladder logic using OpenPLC.
- To create a Start and Stop latching circuit and upload it to Arduino hardware.
- To apply real-world wiring using push buttons and an LED to represent machine control and evaluate the performance and functionality.

2. Materials and Equipments

Arduino UNO	1
Laptop with OpenPLC Editor	1
Light Emitting Diode, LED	1
Push Button	2
Resistor 220 Ω	1
Resistor 10k Ω	2
Breadboard	1
Jumper Wires	

3. Experimental Setup

The experimental setup required both software configuration and hardware wiring. First, OpenPLC Editor has been prepared on the laptop and created a new ladder diagram project. After naming the project, added inputs and outputs as variables for the Start button, Stop button, and LED output. The address of the input and output can be set and follow the OpenPLC physical addressing. Can browse to the physical addressing OpenPLC website for more info regarding the addressing for each type of microcontroller.

Next, we connected the Arduino to the laptop using a USB cable and took note of the COM port number. The circuit on the breadboard used the two push buttons connected as digital inputs, while the LED was connected to a digital output.

The basic flow of the setup:

1. Laptop runs OpenPLC Editor
2. Arduino acts as the PLC runtime
3. Push buttons act as inputs
4. LED acts as output

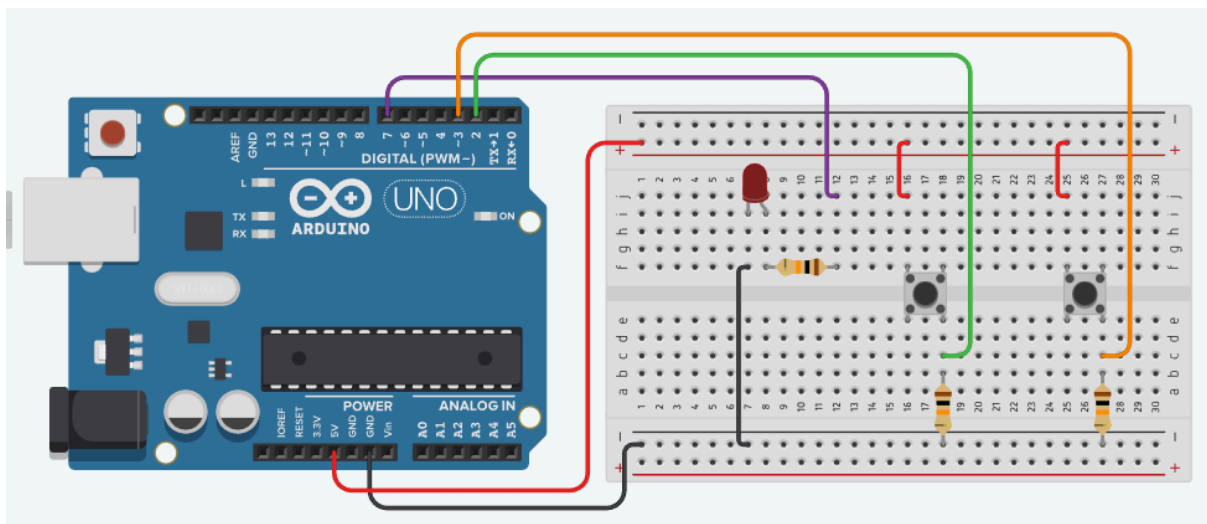


Figure 3.0 Shows the schematic of hardware setup with Arduino using Tinkercad

4. Methodology

4.1. Creating the Ladder Diagram in OpenPLC

1. Open OpenPLC Editor and create a new project.
2. Select Ladder Diagram (LD) as the programming language.
3. Add three variables:
 - StartButton → BOOL, %IX0.0
 - StopButton → BOOL, %IX0.1
 - LED_Output → BOOL, %QX0.0
 - Latch → BOOL
4. Construct the ladder diagram:
 - A Start button normally-open contact, NO
 - A Stop button normally-closed contact, NC
 - A coil for the LED output
 - Use a branch to create a latch (seal-in circuit)

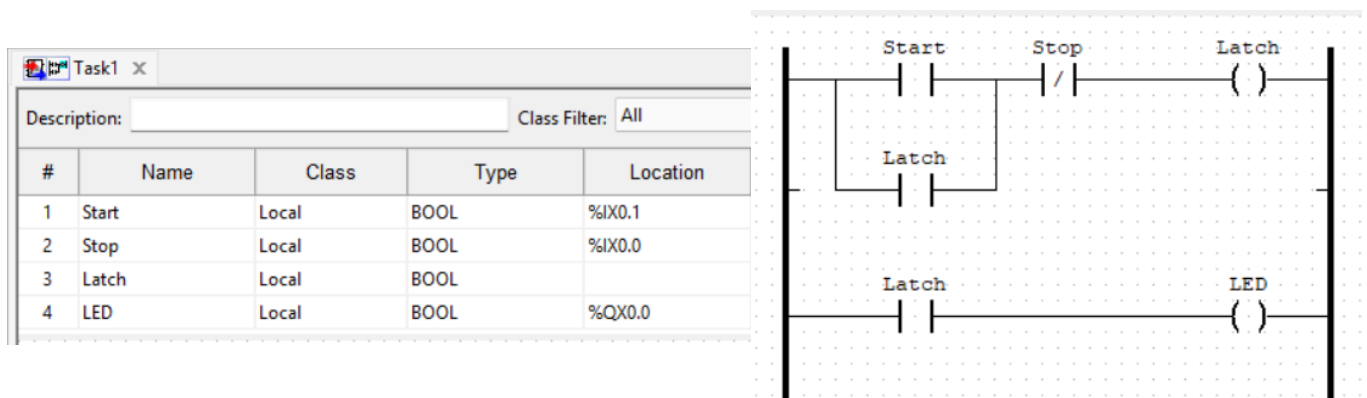


Figure 4.0 Shows the variables and ladder diagram for Start and Stop circuit

4.2. Simulation and Compilation

After building the ladder logic:

1. Click “Compile”
2. Run simulation in the editor
3. Observe LED coil turning ON and OFF when virtual buttons are pressed

This helps verify that the logic works before uploading to the hardware.

4.3 Code Upload to Arduino IDE

1. Connect Arduino to laptop
2. Select the correct COM port
3. Assign correct physical addresses based on Arduino pin mapping
4. Click “Transfer to PLC”
5. Wait for successful upload message

Once uploaded, the Arduino now behaves like a PLC device that follows the ladder logic.

4.4 Building a Physical Circuit

The push buttons and LED were connected on a breadboard as follows:

- Start and Stop Button as Digital Input Pin
- LED as Digital Output Pin

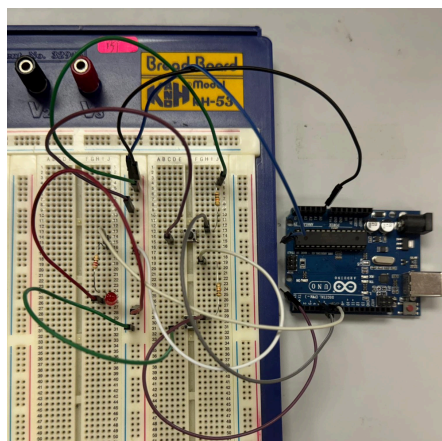


Figure 4.0 Shows the Physical Circuit on breadboard

5. Data Collection

During the real simulation, the Start and Stop buttons were pressed one at a time to observe how the LED reacted. Each action was compared with the expected behavior from a typical industrial Start–Stop circuit. The Start button should switch the LED on, and the LED should remain on even after the button is released because of the latch in the ladder diagram. The Stop button should break the latch and turn the LED off.

The results matched all expectations. Pressing the Start button turned the LED on, releasing the button kept it on, and pressing the Stop button switched it off immediately. Pressing Start again turned the LED back on with no issues. These observations confirmed that the ladder logic, wiring, and Arduino setup performed correctly and consistently. All observations were consistent with normal industrial Start–Stop circuit behavior.

Test	Action	Expected Output	Actual Output
1	Press Start	LED turns ON	LED ON
2	Release Start	LED stays ON	LED stays ON
3	Press Stop	LED turns OFF	LED OFF
4	Press Start	LED turns ON	LED ON

Table 5.0 Shows the observation of the LED

6. Data Analysis

The recorded results show that the program created in the OpenPLC Editor functioned exactly as expected based on the ladder diagram. In the software, the Start and Stop buttons were assigned as **BOOL input variables**, each mapped to their physical addresses according to the OpenPLC addressing system (for example, %IX0.0 for Start and %IX0.1 for Stop, depending on the microcontroller). The LED output was assigned as a **BOOL output variable** (such as %QX0.0).

When the Start button input changed from FALSE to TRUE, the coil in the ladder diagram energized the LED output. The latch branch held the LED output at TRUE because of the internal logic connection, even after the Start input returned to FALSE. This shows that the BOOL flags for both the input and output are updated correctly in the runtime.

The Stop button worked by breaking the latch. When the Stop input changed to TRUE, the logic removed the path that was keeping the LED output energized, so the BOOL output immediately changed to FALSE. This behavior matches standard PLC latch and unlatch control used in automation.

The OpenPLC runtime on the Arduino scanned the program cyclically, updating the input flags, executing the logic, and refreshing the output flags each cycle. Because the LED output responded instantly and consistently, the analysis confirms that both the software addressing and the hardware inputs/outputs were mapped correctly and that the PLC scan cycle handled the BOOL states without delay.

7. Results

The results confirm that the OpenPLC program and the Arduino hardware responded exactly as intended. The Start button's BOOL input successfully energized the LED's BOOL output, and the latch in the ladder diagram kept the output TRUE even after the Start input returned to FALSE. Pressing the Stop button changed its input flag to TRUE, which broke the latch and switched the LED output back to FALSE.

Both the OpenPLC simulation and the real Arduino setup produced the same behavior, showing that the input and output addresses were mapped correctly and that the PLC scan cycle updated all BOOL states reliably. The experiment demonstrated a stable and fully working Start–Stop control system.

Test	Action	Output
1	Press Start	LED ON
2	Release Start	LED stays ON
3	Press Stop	LED OFF
4	Press Start	LED ON

Table 7.0 Shows the Results from the LED

8. Discussion

The experiment successfully demonstrated how a PLC-style Start–Stop control circuit can be implemented using OpenPLC Editor and an Arduino microcontroller. The results observed from both simulation and hardware testing confirmed the correct operation of the ladder logic, proving that the latch and unlatch mechanism works reliably when mapped into a microcontroller environment.

One important finding from this experiment is how closely the OpenPLC runtime mimics the behaviour of actual industrial PLCs. Even though the hardware used was an Arduino UNO, the scan-cycle execution (reading inputs → processing logic → updating outputs) functioned exactly like a real PLC. This was evident from the consistent timing and immediate response of the LED after each button press.

The experiment also highlighted the importance of proper physical addressing. The mapping of variables such as `%IX0.0`, `%IX0.1`, and `%QX0.0` must match the actual digital pins on the Arduino. Any mismatch would cause incorrect input readings or no LED output. Ensuring that the OpenPLC variables corresponded to the correct pins was crucial for achieving accurate results.

From a hardware perspective, the wiring of the push buttons required attention because incorrect pull-up or pull-down resistor configurations could lead to unstable input states, such as bouncing or false triggering. Using the 10 kΩ resistors ensured that the inputs remained stable when the buttons were not pressed. This stability allowed the ladder logic to function without unexpected behaviour during testing.

Overall, the experiment demonstrated that combining PLC logic with a microcontroller provides a flexible and cost-effective way to learn industrial automation concepts. It allowed us to explore how ladder logic behaves in real-world hardware and provided a hands-on understanding of latching circuits, boolean logic, and input-output mapping.

9. Conclusion

This laboratory experiment successfully achieved its objectives by demonstrating the implementation of a Start–Stop control circuit using OpenPLC and an Arduino microcontroller. We learned how to construct ladder logic, assign variables, map physical addresses, and test the program through simulation and real hardware.

The LED responded correctly in all tests, proving that the latch and unlatch logic worked as expected. The Start button activated the output and held it ON through the latch mechanism, while the Stop button successfully broke the latch and turned the LED OFF. The behaviour in the simulation matched the real hardware output, confirming the accuracy of both the wiring and software configuration.

Through this experiment, we gained practical knowledge in PLC fundamentals, including scan cycles, boolean logic control, digital input and output handling, and the importance of proper addressing. This experience serves as a strong foundation for more complex automation systems, where more inputs, outputs, and control conditions may be introduced.

Overall, the experiment demonstrated the reliability and simplicity of PLC-style control using OpenPLC and provided valuable hands-on exposure to industrial automation concepts.

10. Recommendations

Based on the completion of this experiment, several improvements can be considered to enhance the reliability of the system and the overall learning experience:

1. Ensure Stable Push Button Inputs

Adding simple hardware debouncing using resistors or capacitors can help stabilise the push button signals. This prevents unintentional multiple triggers and ensures that the Start and Stop commands behave consistently.

2. Maintain Organised Wiring Layout

Arranging the breadboard wiring neatly and avoiding tangled jumper wires can make troubleshooting easier. A clean layout helps in identifying incorrect connections and ensures the circuit functions as expected.

3. Double-Check Physical Addressing Before Uploading

Verifying the assigned input and output addresses according to the selected Arduino pins helps avoid errors during runtime. A quick check reduces issues such as the LED not responding or inputs not being detected.

4. Test Input and Output Devices Individually

Before running the full ladder logic program, testing the push buttons and LED using simple `digitalRead` and `digitalWrite` behaviour can help confirm that the hardware is functioning correctly.

5. Introduce Timer or Interlock Elements Gradually

In future experiments, adding basic timer blocks or interlocks can help strengthen understanding of more advanced PLC functions while still keeping the circuit simple.

These recommendations aim to support smoother execution of the experiment while improving the accuracy and stability of both the software and hardware aspects of the PLC–microcontroller system.

11. References

OpenPLC. “2.4 Physical Addressing – Autonomy.” *Autonomy*,
autonomylogic.com/docs/2-4-physical-addressing/.

“Autonomy – Open-Source PLC Software.” *Autonomy*, autonomylogic.com/.

Wilcher, Dr Don. “PLC Ladder Logic on an Arduino: Building a Start-Stop Circuit.”
Control.com, 28 Sept. 2022,
control.com/technical-articles/plc-ladder-logic-on-an-arduino-building-a-start-stop-circuit/.

“PLC Ladder Logic on an Arduino: Introduction to OpenPLC.” *Control.com*, 30 Aug. 2022,
control.com/technical-articles/plc-ladder-logic-on-an-arduino-introduction-to-openplc/.

Appendix A - Circuit Diagram

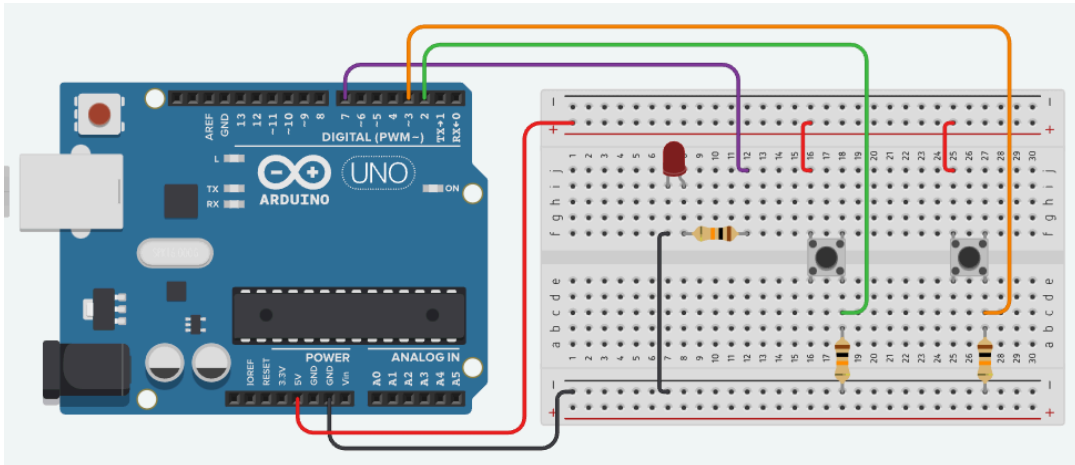


Figure A1 Shows the connection of circuit configuration by using Tinkercad

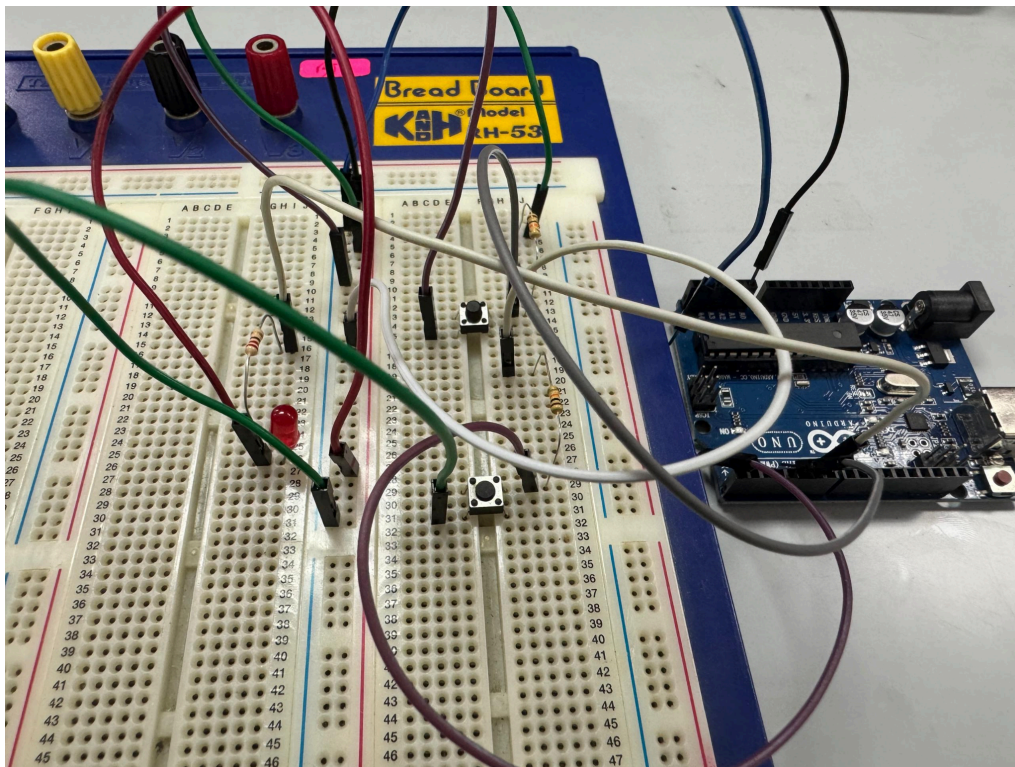


Figure A2 Shows the real connection of circuit hardware

Appendix B - Ladder Diagram in OpenPLC

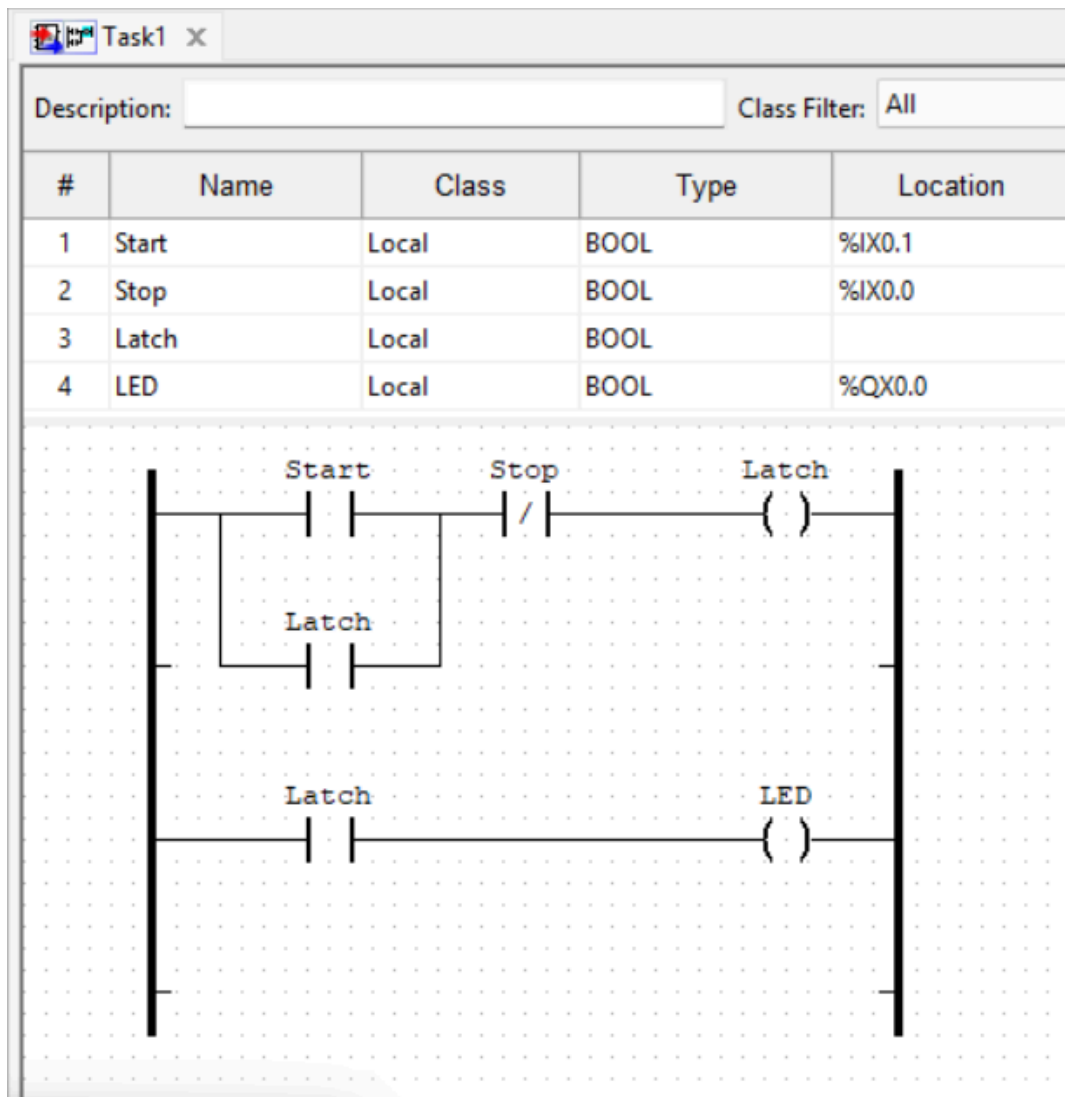


Figure B Shows the setup in the OpenPLC

Appendix C - Results

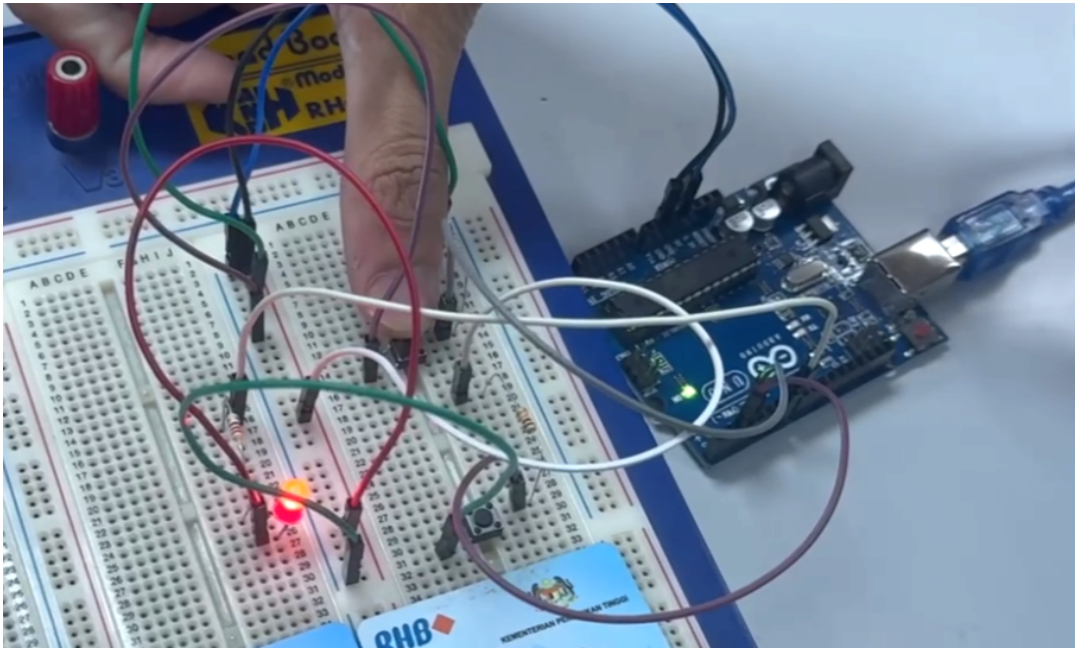


Figure C1 Shows the output when Start PushButton is TRUE

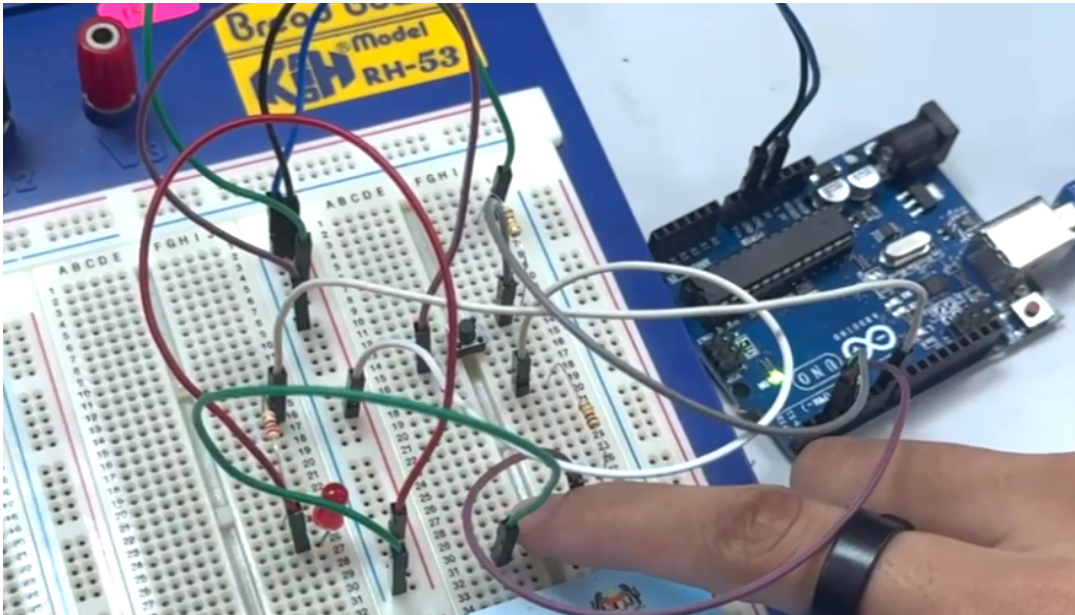


Figure C2 Shows the output when Stop PushButton is TRUE

Acknowledgments

We would like to express our sincere appreciation to our lecturer, Dr. Zulkifli, and lab assistant for his guidance and clear explanations throughout this experiment. His teaching on PLC systems and the integration of Arduino with OpenPLC helped us understand the concepts more easily. His support during the lab session also made it easier to configure the software, assign input and output addresses, and verify the operation of the Start–Stop circuit.

We are also thankful to the lab assistants for assisting with the equipment setup, wiring checks, and troubleshooting. Their help ensured that the hardware connections were safe, stable, and functioning correctly before running the program. Their guidance played an important role in completing the circuit smoothly.

We appreciate the effort and cooperation of all group members. Each member contributed to assembling the circuit on the breadboard, preparing the ladder logic in OpenPLC Editor, testing the input and output responses, and completing the report. Working as a team made the whole process more efficient and enjoyable.

Lastly, we would like to thank our classmates for sharing ideas, components, and offering help whenever needed during the lab session. The collaborative environment created by everyone contributed to the successful completion of this experiment. Overall, we are truly grateful to our lecturer, lab assistants, teammates, and classmates for their support and guidance throughout this lab.

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specific in references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or a person.

We hereby certify that this report has **not been done by only one individual and all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We, therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us**.

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Name: Muhammad Fahim Bin Ahmad Norisham

Matric Number: 2312369

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Name: Muhammad Zamarul Azim Bin Zulkhibri

Matric Number: 2312451

Read /

Understand /

Agree /

Signature: 

Name: Muhammad Hazimuddin Bin Fairuz

Matric Number: 2312479

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